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Referee Report
Paper: "Dynamic Spatial Competition in Early Education: An Equilibrium Analysis of the Preschool Market in Pennsylvania"
Author: Pierre Bodéré
Recommendation: Revise and Resubmit

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1. Summary

This paper develops a dynamic model of the preschool market in Pennsylvania. It combines a static demand model of parents' choices with a pricing model for Early Childhood Education (ECE) providers, and then embeds this in a dynamic framework where providers decide whether to enter, exit, or invest in quality. The paper solves the dynamic game by using parametric policy iteration with neural networks to approximate variable profits across states, which makes the computation tractable.

The model is estimated using Pennsylvania ECE provider data from 2010–2018. The data provides information on the quality, enrollment and subsidy status of each child. The data is then used to evaluate different policy designs by comparing counterfactuals; the paper mainly focuses on 3 policy proposals. The paper looks at policies that expand subsidies to middle-income families and policies that give targeted financial support to ECE providers; both of these policies work through the static portion of the model. One more policy the paper focuses on is the impact of start-up grants, which works through the dynamic decision model of the preschools.

The main finding is that demand subsidies to lower-income families, combined with targeted financial support for high-quality providers serving low-income children(ie, add-on rates), are more cost-effective and preserve quality more than broad demand subsidies or entry subsidies.

2. General Remarks

This is a thorough paper that studies an important policy question. It establishes a strong argument for why ECE and the quality of it are so important and how the US has a concerning lack of it. The paper does a good job of explaining the trade-offs for different types of policy. That broad subsidies to families can increase the price for parents who are not eligible, but broad entry subsidies for ECE providers can lead to a decrease in quality.

A key strength is that it combines short-run pricing decisions with longer-run entry and investment decisions in a single framework. This allows the paper to measure both immediate and longer-term responses and effects to policy changes, which differentiates it from related literature.

The empirical setting is also well chosen. The Pennsylvania data is very detailed and makes this type of analysis possible in a way that would be difficult in most other contexts.

Overall, I think the paper makes a meaningful contribution and is worth revising. My main concerns relate to whether the model captures behaviour in a realistic enough way to support the policy conclusions.

3. Main Suggestions

Comment 1. Substitution patterns in demand

The paper incorporates the importance of the location of the preschools by defining geographic markets and including distance directly in household utility. However, because the geographic markets are still quite large, and the demand model uses a standard logit structure within these markets, substitution patterns may still be restricted.

In particular, distance affects utility level, but the model may not fully capture the idea that, conditional on market shares and utility, two nearby centers are closer substitutes for each other relative to two centers that are farther apart. This matters for the counterfactuals because entry or subsidy policies may have highly local competitive effects. Therefore, it would be useful to report substitution patterns by distance or to consider adding smaller geographic nests.

Comment 2. Instruments and possible demand changes

The paper uses competitors' characteristics and local market conditions as instruments for price. Appendix E explains this more and says the instruments are based on the Gandhi-Houde approach. It also says that the final instruments are chosen from a much larger set of possible instruments.

One concern is that, since the model is dynamic, providers might be responding to changes in neighbourhoods over time. For example, if a neighbourhood is becoming richer or more attractive to families, centers may enter, upgrade quality, or change prices before these changes are fully captured in the data. If this happens, then the instruments may be related to unobserved demand changes, which would weaken the identification strategy.

The paper partially addresses this concern through the timing of the dynamic game. Since firms make entry and quality decisions before the transitory shock is realized, competitors' characteristics in that period are considered unrelated to that period's transitory shock. But this doesn't fully resolve it. If firms are forward-looking, their entry and upgrade decisions will already reflect what they expect demand to look like in the future. Competitors' characteristics would then be picking up anticipated demand conditions, not just costs. The paper has evidence that shows that upgrade and exit decisions differ across income and education levels, so a robustness check using a more restricted instrument set, such as physical capacity or distance-based measures, would help address this.

Second, the paper picks the final instruments after looking at which ones work best in the first stage. This is not necessarily wrong, but it makes the first-stage results a little less convincing. Since the author starts with many possible instruments, some of them might look strong just because they fit this sample well. It would be helpful to show that the main results still hold when using a smaller set of instruments chosen for clear economic reasons, not just because they perform well in the first stage.

Comment 3. Behaviour of non-profit providers

The paper assumes that all providers behave as profit-maximizing firms. The author provides evidence that centers respond to financial incentives (for example, higher subsidies increase the probability of upgrading quality), which is consistent with profit-maximizing behaviour.

This evidence might not be enough to fully justify the assumption, especially for non-profit providers. Non-profits may still respond to financial incentives while also placing more weight on other aspects, like enrollment levels or quality provision. The paper acknowledges this issue in footnote 24 but doesn't offer any robustness check to assess how much it matters.

This matters because the model uses profit maximization as a basis for all pricing behaviour. If non-profits have different objectives, the model may attribute their pricing decisions to differences in costs rather than differences in preferences. This could affect the estimated markups and, in turn, the policy counterfactuals.

As a robustness check, it would be helpful to estimate the pricing model separately by ownership type or to test whether the implied markups for non-profits are consistent with profit-maximizing behaviour.

4. Minor Improvements

The paper uses a discount factor of 0.9 without explaining why this value was chosen or how sensitive the results are to it. Since the discount factor affects how providers weigh future profits, and the conclusions depend on providers' dynamic decisions, it would be worth checking that the results hold under other values or justify why 0.9 was chosen.

5. Conclusion

Overall, this is a thoughtful and well-executed paper on an important policy question. The paper's main contribution is that it combines parents' choices and provider pricing with longer-run provider decisions, such as entry, exit, and quality investment. This is useful because ECE policies don't only affect prices and enrollment in the short run, but can also change the type and quality of providers in the market over time.

My main concerns are about whether some of the assumptions in the model are strong enough to fully support the counterfactual results. In particular, the substitution patterns between centers, the instrument strategy, and the behaviour of non-profit providers all matter for how we interpret the policy simulations. These issues do not take away from the main contribution of the paper, but addressing them would make the policy conclusions more convincing.

I recommend revising and resubmitting. The paper focuses on an important policy setting using strong data and makes a meaningful contribution. I think with some additional robustness checks and a clearer discussion of the main assumptions, the paper would be stronger.